

52-amaliy mashg'ulot

Na'muna uchun misollar yechimi

1-misol. Ushbu

$$f(x, y) = \arctg \frac{x}{y} \quad (y \neq 0)$$

funksiyaning ikkinchi tartibli xususiy hosilalari topilsin.

◀ Ravshaki,

$$\frac{\partial f}{\partial x} = \frac{1}{1 + \frac{x^2}{y^2}} \cdot \frac{1}{y} = \frac{y}{x^2 + y^2}, \quad \frac{\partial f}{\partial y} = \frac{1}{1 + \frac{x^2}{y^2}} \cdot \left(-\frac{x}{y^2} \right) = -\frac{x}{x^2 + y^2}$$

bo'ladi.

Endi ta'rifdan foydalanib berilgan funksiyaning ikkinchi tartibli xususiy hosilalarini topamiz:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{y}{x^2 + y^2} \right) = -\frac{2xy}{(x^2 + y^2)^2},$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2} \right) = \frac{x^2 - y^2}{(x^2 + y^2)^2},$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(-\frac{x}{x^2 + y^2} \right) = -\frac{x^2 - y^2}{(x^2 + y^2)^2},$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(-\frac{x}{x^2 + y^2} \right) = \frac{2xy}{(x^2 + y^2)^2}. \quad \blacktriangleright$$

2-misol. Ushbu

$$f(x,y)=\begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & \text{agar } x^2 + y^2 > 0 \text{ бўлса,} \\ 0, & \text{agar } x^2 + y^2 = 0 \text{ бўлса} \end{cases}$$

funksiyaning $(0,0)$ nuqtadagi aralash hosilalari topilsin.

Aytaylik, $(x,y) \neq (0,0)$ bo'lsin. Bu holda

$$\frac{\partial f}{\partial x} = y \left(\frac{x^2 - y^2}{x^2 + y^2} + \frac{4x^2 y^2}{(x^2 + y^2)^2} \right), \quad \frac{\partial f}{\partial y} = x \left(\frac{x^2 - y^2}{x^2 + y^2} - \frac{4x^2 y^2}{(x^2 + y^2)^2} \right),$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{x^2 - y^2}{x^2 + y^2} \cdot \left(1 + \frac{8x^2 y^2}{(x^2 + y^2)^2} \right),$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{x^2 - y^2}{x^2 + y^2} \cdot \left(1 + \frac{8x^2 y^2}{(x^2 + y^2)^2} \right)$$

bo'ladi.

Aytaylik, $(x,y) = (0,0)$ bo'lsin. Bu holda funksiyaning hosilalarini ta'rifga ko'ra hisoblaymiz:

$$\frac{\partial f(0,0)}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x} = 0,$$

$$\frac{\partial f(0,0)}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0,0)}{\Delta y} = 0,$$

$$\frac{\partial^2 f(0,0)}{\partial y \partial x} = \lim_{\Delta y \rightarrow 0} \frac{\frac{\partial f(0, \Delta y)}{\partial x} - \frac{\partial f(0,0)}{\partial x}}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{-\Delta y^3}{\Delta y^3} = -1,$$

$$\frac{\partial^2 f(0,0)}{\partial x \partial y} = \lim_{\Delta x \rightarrow 0} \frac{\frac{\partial f(\Delta x, 0)}{\partial y} - \frac{\partial f(0,0)}{\partial y}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{-\Delta x^3}{\Delta x^3} = -1 \quad \blacktriangleright$$

Misollar

1. Ushbu

$$u = f(x, y), \quad x = t^2 + s^2, \quad y = t \cdot s$$

funksiyaning ikkinchi tartibli differensialini topilsin.

2. Ushbu

$$u = y \cdot \varphi(x^2 - y)$$

funksiya quyidagi

$$\frac{1}{x} \frac{\partial u}{\partial x} + \frac{1}{y} \frac{\partial u}{\partial y} = \frac{u}{y^2}$$

tenglikni qanoatlantirishi isbotlansin.

3. Quyidagi funksiyalarning ikkinchi tartibli differensiallari topilsin.

1. $z = 2x^2 - 3xy - y^2$

2. $z = e^{xy}$

3. $z = \frac{x}{y}$

4. $z = \ln \sqrt{x^2 + y^2}$

5. Agar $f(x, y) = x^2 + xy + y^2 - 4 \ln x - 10 \ln y$ bo'lsa, $d^2 f(1, 2)$ topilsin.

6. $z = y \ln x$

7. $z = \sin x \cos y$